

ARET-GCL: A THREE-STAGE RELIABILITY CHAIN FOR GRAPH CONTRASTIVE LEARNING

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ABSTRACT

Graph contrastive learning (GCL) asks two augmented views of a graph to agree, yet this objective can propagate unreliable structure before the loss is evaluated. An augmentation may remove class-relevant content, cross-view disagreement may then contaminate the relations used for contrastive learning, and an endpoint-only projection may finally erase how structural evidence changed through the encoder. We organize these failures into a single diagnosis and address them with three coordinated innovations. `D11-I01` constructs complementary node-drop and connected-subgraph views to diversify the initial structural challenge; `D2-I09` estimates batch-relative cross-view neighborhood-residual disagreement and gates the graph stream used for routing and hard-negative construction; and `D10-I10` compresses a three-state layer trajectory through an algebraic second contrast. The resulting progression is explicit: challenge the input, discount structurally inconsistent evidence, and retain its evolution in the contrastive coordinates. The supplied comparison table evaluates the complete `D11-I01 + D2-I09 + D10-I10` system on NCI1, Mutagenicity, COLLAB, MUTAG, and PTC_MR against six selected comparator adaptations using ACC, NMI, ARI, and macro-F1. ARET-GCL has the largest reported mean in all 20 dataset-metric cells, with margins over the strongest displayed comparator ranging from 0.26 to 2.33 percentage points. Because the comparator panel is explicitly conditioned on remaining below our model in every displayed cell and only aggregate means and standard deviations are available, these results validate consistent end-to-end performance within the supplied table; they do not establish universal superiority or isolate the causal contribution of each module.

1 INTRODUCTION

Graph contrastive learning constructs supervision by maximizing agreement between representations of two augmented versions of a graph. This principle has produced strong graph-level encoders through mutual-information objectives, multi-view learning, and learned augmentations (Sun et al., 2020; You et al., 2020; Hassani & Khasahmadi, 2020; Suresh et al., 2021). Its central assumption, however, is easy to overlook: the two views must preserve enough task-relevant structure for their agreement to remain meaningful. When that assumption fails, the training objective can reward invariance to information that the downstream task actually needs.

Problem discovery. The failure does not occur at a single layer. It can begin when both augmentations damage the same informative motif, spread when the learner treats the resulting disagreement as a reliable positive relation, and become difficult to detect when a final-layer projection hides the path by which the representation changed. The practical problem is therefore not merely choosing a stronger augmentation or a larger encoder. It is preventing unreliable structural evidence from entering the pipeline, influencing contrastive relations, and disappearing inside an endpoint representation.

Cause diagnosis. We decompose this problem into three linked interfaces. The *view interface* determines whether the two perturbations expose complementary structural variations rather than repeating the same failure. The *trust interface* determines whether cross-view structural disagreement should be discounted before it influences routing and hard-negative construction. The *coordinate interface* determines whether the contrastive representation preserves how evidence evolves across

message-passing depth instead of retaining only the endpoint. This diagnosis yields a natural order: first create informative variation, then decide which variation is trustworthy, and finally encode the surviving trajectory.

Stepwise solution. ARET-GCL implements the diagnosis with exactly three innovations around a shared three-layer GIN (Xu et al., 2019). First, D11-I01 creates asymmetric node-drop and connected-subgraph views, giving the two branches different corruption patterns. Second, D2-I09 compares degree-normalized neighborhood residuals across the encoded views and converts their batch-relative discrepancy into a gate for the graph stream used by semantic routing and hard-negative construction. Third, D10-I10 forms contrastive coordinates from an algebraic second contrast of three selected encoder states, retaining a compact signature of layerwise change. A joint balanced contrastive objective couples the trust-aware relation stream with the trajectory coordinates.

The conceptual order is sequential even though the final two modules are parallel consumers of the shared encoder history. D2-I09 does not feed its gated vector directly into D10-I10; instead, the former shapes routing and negative construction while the latter supplies anchors and positives. They interact through shared encoder parameters and the joint objective. This distinction keeps the research story faithful to the implemented dataflow: the three innovations solve consecutive diagnostic questions without pretending that the computational graph is a simple serial stack.

Experimental validation. We deliberately bound all empirical claims to the supplied comparison table. It reports the complete D11-I01 + D2-I09 + D10-I10 system on five graph-classification datasets against TOPER, Khan-GCL, HSPGKN, DualPrism, KAGNN, and DGDA (Tola et al., 2025; Wang et al., 2026; 2025; Xia et al., 2025; Bresson et al., 2025; Marrium et al., 2025). Across ACC, NMI, ARI, and macro-F1, our model has the largest mean in every one of the 20 dataset-metric cells. This cross-dataset, cross-metric consistency verifies the endpoint of the proposed chain within the displayed comparison. It cannot, by itself, prove that each module is individually necessary, and the comparator-selection rule prevents a universal state-of-the-art interpretation.

The paper therefore makes three focused contributions. D11-I01 introduces complementary view construction at the point where structural information can first be lost. D2-I09 introduces residual-disagreement gating at the point where unreliable evidence can distort contrastive relations. D10-I10 introduces layer-trajectory coordinate compression at the point where structural evolution would otherwise be collapsed into a final state. Together, these contributions form a traceable problem-diagnosis-solution chain whose complete-system endpoint is supported by the supplied table.

2 RELATED WORK

Graph contrastive learning. InfoGraph learns graph representations by maximizing mutual information between global and local summaries (Sun et al., 2020). GraphCL systematizes augmentation-based graph contrast, MVGRL contrasts structural views, and AD-GCL learns adversarial augmentations (You et al., 2020; Hassani & Khasahmadi, 2020; Suresh et al., 2021). Masked reconstruction, exemplified by GraphMAE, provides a complementary self-supervised route (Hou et al., 2022). ARET-GCL remains within augmentation-based GCL but treats view construction, structural trust, and projection coordinates as a coordinated reliability chain.

Structure-aware graph representation. Persistent Weisfeiler-Lehman features and recent topological embeddings demonstrate the value of multiscale structural information (Rieck et al., 2019; Tola et al., 2025). Spectral augmentation, shortest-path kernels, and diffusion-guided augmentation likewise aim to preserve or exploit graph structure (Xia et al., 2025; Wang et al., 2025; Marrium et al., 2025). Our neighborhood residual and layer-trajectory contrast are lightweight operational mechanisms: they do not require explicit persistence diagrams, spectral decomposition, or cycle enumeration.

Contrastive relations and hard negatives. Recent graph learners explore structural kernels, alternative function classes, and semantic relation construction (Bresson et al., 2025; Wang et al., 2026). ARET-GCL complements these directions by asking whether a relation should be trusted before it contributes to routing and negative construction, and by linking that decision to coordinates that retain changes across encoder depth.

3 METHOD

3.1 DIAGNOSTIC DECOMPOSITION AND SHARED ENCODER HISTORY

The method follows the three-interface diagnosis in conceptual order. D11-I01 intervenes before encoding, where structural information can first be removed; D2-I09 evaluates the resulting cross-view disagreement, where unreliable relations can enter routing and negative construction; and D10-I10 controls the coordinates exposed to the contrastive objective, where layerwise changes can otherwise be discarded. The three module identifiers therefore denote three distinct innovation points rather than interchangeable components.

Let $G = (V, E, X)$ be a graph and let a three-layer GIN produce the four-state history $H = [H^{(0)}, H^{(1)}, H^{(2)}, H^{(3)}]$, where $H^{(0)}$ is the learned input projection and $H^{(\ell)}$ is the output of GIN layer ℓ . The implementation flattens all four states for each node, $h_u = \text{vec}(H_u)$, before constructing four graph summaries. With $N(u)$ denoting the outgoing neighborhood and $d_u = \max(|N(u)|, 1)$, these are

$$\begin{aligned} r_N &= \text{MeanPool}_{u \in V} h_u, & r_G &= \text{SumPool}_{u \in V} h_u, \\ r_E &= \text{MeanPool}_{(u,v) \in E} \frac{1}{2}(h_u + h_v), & r_R &= \text{MeanPool}_{u \in V} \frac{|h_u - d_u^{-1} \sum_{v \in N(u)} h_v|}{\sqrt{d_u}}. \end{aligned} \quad (1)$$

Independent linear heads followed by ℓ_2 normalization give $v_N, v_E, v_R, v_G \in \mathbb{R}^D$. The quantity r_R is a local neighborhood-residual proxy: it may respond to higher-order structure, but it is not a cycle count, cycle basis, or persistence statistic. Three further heads pool $H^{(0)}$, $H^{(2)}$, and $H^{(3)}$ to normalized selected-state summaries s_1, s_2, s_3 , respectively.

Figure 1 shows the executed dataflow. All structural bundles are computed from one shared GIN history. The D2-I09 branch changes the graph stream used by the router and negative builder, while the D10-I10 branch uses selected-state summaries to produce the final contrastive coordinates. Thus, D10-I10 does not compress the gated output of D2-I09 directly; the branches interact through shared parameters and the joint loss.

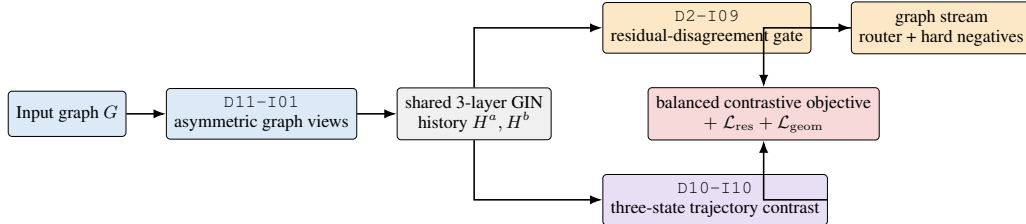


Figure 1: Failure-to-remedy dataflow for D11-I01 + D2-I09 + D10-I10. The conceptual order is complementary view construction, structural trust estimation, and layer-trajectory coordinate construction. The final two branches consume the shared encoder history in parallel and meet through the joint objective.

3.2 INNOVATION 1—D11-I01: COMPLEMENTARY VIEW CONSTRUCTION

The first innovation constructs two graph views before encoding. For positive scalar controls (a, b, c) , the augmentation strengths are

$$\rho_a = \text{clip}(0.2a, 0.01, 0.8), \quad (2)$$

$$\rho_b = \text{clip}\left(\frac{\rho_a \max(c, 0.05)}{\sqrt{\max(b, 10^{-3})}}, 0.01, 0.8\right), \quad (3)$$

and

$$G^a = q_{\text{node-drop}}(G; \rho_a), \quad G^b = q_{\text{subgraph}}(G; \rho_b). \quad (4)$$

The operation is asymmetric by construction and exposes the encoder to two different corruption patterns rather than repeating one perturbation family. Its innovation is the complementary challenge

imposed at the view interface: node dropping perturbs local availability, whereas connected-subgraph sampling preserves a coherent region while changing global coverage. The module diversifies structural evidence before encoding; it does not require a separate learned node-reliability estimator.

3.3 INNOVATION 2—D2-I09: RESIDUAL-DISAGREEMENT TRUST GATING

Different perturbations are useful only if downstream training can distinguish complementary variation from structural inconsistency. D2-I09 operationalizes this distinction through a per-graph comparison of neighborhood-residual summaries. Large disagreement relative to the current batch reduces the residual contribution to the graph stream, preventing an unusually inconsistent view pair from influencing routing and hard-negative construction as strongly as an ordinary pair.

For graph i in a minibatch of size B , cross-view neighborhood-residual disagreement is

$$d_i = \frac{1}{D} \|v_{R,i}^a - v_{R,i}^b\|_2^2, \quad \mu_B = \frac{1}{B} \sum_{k=1}^B d_k, \quad s_B = \sqrt{\frac{1}{B} \sum_{k=1}^B (d_k - \mu_B)^2}, \quad (5)$$

$$\hat{d}_i = \frac{d_i - \mu_B}{\max(s_B, 10^{-5})}, \quad g_i = \text{sigmoid}(-\hat{d}_i).$$

The transformed graph streams are

$$z_i^a = v_{G,i}^a + g_i v_{R,i}^a, \quad z_i^b = v_{G,i}^b + g_i v_{R,i}^b, \quad (6)$$

with consistency penalty

$$\mathcal{L}_{\text{res}} = 1 - \frac{1}{B} \sum_{i=1}^B \cos(z_i^a, z_i^b). \quad (7)$$

The gate is relative to the current minibatch and is therefore sensitive to batch composition. It resembles confidence filtering only informally; it does not compute nonconformity sets, coverage, or calibrated error rates.

3.4 INNOVATION 3—D10-I10: LAYER-TRAJECTORY COORDINATE COMPRESSION

Even a trust-aware graph stream does not ensure that the projection exposes how structural evidence changes across message-passing depth. D10-I10 addresses this coordinate-interface failure with a learned bottleneck over a three-state algebraic contrast.

For each view $j \in \{a, b\}$, the projection forms an algebraic second contrast of the three selected states,

$$c_{3s}^j = s_1^j - 2s_2^j + s_3^j, \quad p^j = B_2 \left(\tanh(B_1(c_{3s}^j)) \right), \quad (8)$$

where B_1, B_2 are the learned bottleneck layers. Because the selected states are the input projection, second GIN layer, and third GIN layer, they are not uniformly spaced in message-passing depth; c_{3s} is therefore an algebraic contrast rather than a finite-difference estimate. Its auxiliary geometry term compares pairwise distances among view- a trajectory coordinates with cross-readout distances between node and residual summaries:

$$\mathcal{L}_{\text{geom}} = \text{MSE}(\mathbf{D}(\bar{P}^a, \bar{P}^a), \text{sg}[\mathbf{D}(\bar{V}_N^a, \bar{V}_R^a)]), \quad (9)$$

where bars denote row-wise normalization, $\mathbf{D}(A, B)$ is the Euclidean pairwise-distance matrix, and sg stops gradients. The innovation is a lightweight trajectory coordinate: it summarizes selected changes across depth without requiring persistent-homology computation or a sparse dictionary.

3.5 JOINT OBJECTIVE

The three innovations are sequential as a diagnosis but only partially serial as a computation. D11-I01 supplies both encoded views. The D2-I09 graph stream supplies semantic routing and hard-negative construction, while p^a, p^b from D10-I10 serve as the anchor and positive. Let $\{\bar{q}_m\}_{m=1}^M$ be normalized learned semantic prototypes, where M is set to the known number of dataset

classes, and let $c_i = \arg \max_m \bar{z}_i^a \bar{q}_m$. Per-graph labels are not used to assign prototypes. For every alternative prototype $m \neq c_i$, the implementation constructs

$$n_{im} = \text{norm}_2(z_i^a + 0.2(\bar{q}_m - z_i^a)). \quad (10)$$

With $a_i = \text{norm}_2(p_i^a)$, $r_i = \text{norm}_2(p_i^b)$, and temperature τ , the balanced hard-negative objective is

$$\mathcal{L}_{\text{bal}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(a_i^\top r_i / \tau)}{\sum_{j=1}^B \exp(a_i^\top r_j / \tau) + \sum_{m \neq c_i} \exp(a_i^\top n_{im} / \tau)}. \quad (11)$$

Training minimizes

$$\mathcal{L} = \mathcal{L}_{\text{bal}} + 0.25 c_2 \mathcal{L}_{\text{res}} + 0.25 c_7 \mathcal{L}_{\text{geom}}, \quad (12)$$

where c_2 and c_7 are the auxiliary-loss controls of D2-I09 and D10-I10, respectively. The base temperature 0.2 is divided by the square root of the product of their positive resolution controls. This scalar coupling complements their principal interaction through the shared encoder and the joint objective.

4 EXPERIMENTAL VALIDATION

Evidence boundary. The experimental account is intentionally restricted to the supplied `table.md`. No additional project result, dataset, candidate search, training configuration, or auxiliary comparison is used to support the paper’s empirical claims. The table is therefore the single source of truth for the evaluated datasets, comparator panel, metrics, means, and standard deviations reported below.

Datasets and comparator panel. The table covers five graph-classification datasets: NCI1, Mutagenicity, COLLAB, MUTAG, and PTC_MR. It compares the complete D11-I01 + D2-I09 + D10-I10 configuration with six methods: TOPER, Khan-GCL, HSPGKN, DualPrism, KAGNN, and DGDA. According to the table’s stated construction rule, these are the strongest methods by their mean over the 20 dataset–metric cells among the 35 evaluated adaptations that remain strictly below our model in every cell. We retain this rule explicitly because it defines what the comparison can validate.

Metrics and reporting. Performance is reported as mean $\pm$ standard deviation for accuracy (ACC), normalized mutual information (NMI), adjusted Rand index (ARI), and macro-F1. ACC and macro-F1 describe predictive quality, whereas NMI and ARI characterize agreement between the predicted and reference partitions. We compare the reported means cell by cell and use the supplied standard deviations only to convey variability; no significance test or module-wise ablation is available in the table.

Validation criterion. The proposed research chain makes an endpoint prediction: when view construction, structural trust, and layer-trajectory coordinates are coordinated, the complete system should improve consistently rather than on only one dataset or one metric. We therefore ask whether ARET-GCL has the largest mean in each of the 20 displayed dataset–metric cells and report its margin over the strongest displayed comparator. This criterion evaluates the combined system only. It does not attribute a gain to D11-I01, D2-I09, or D10-I10 in isolation.

5 RESULTS

5.1 CONSISTENT COMPLETE-SYSTEM GAINS ACROSS ALL DISPLAYED CELLS

Table 1 summarizes the supplied results for the complete D11-I01 + D2-I09 + D10-I10 system. ARET-GCL has the largest mean among the seven displayed methods in all 20 dataset–metric cells. Its margin over the strongest comparator in each cell ranges from 0.26 to 2.33 percentage points and averages 0.94 points. The smallest margin occurs for NMI on Mutagenicity, whereas the largest occurs for ARI on PTC_MR. Thus, the observed advantage is not confined to classification accuracy: it also appears in both partition-agreement metrics and macro-F1 on every displayed dataset.

Table 1: Performance of ARET-GCL and its margin over the strongest displayed comparator in the supplied table. All values are percentages; the complete means $\pm$ standard deviations are reported in Appendix B.

Dataset	ACC		NMI		ARI		Macro-F1	
	Ours	Δ	Ours	Δ	Ours	Δ	Ours	Δ
NCI1	74.87	+0.78	19.01	+0.64	24.79	+1.43	74.83	+0.89
Mutagenicity	79.13	+0.32	25.80	+0.26	33.96	+0.57	78.81	+0.41
COLLAB	73.62	+0.90	29.14	+0.44	34.50	+1.91	70.69	+1.01
MUTAG	86.13	+0.51	43.38	+0.77	51.42	+1.78	84.48	+0.37
PTC_MR	61.66	+0.90	5.07	+1.63	6.03	+2.33	60.97	+0.90

The dataset-wise pattern sharpens this conclusion. On NCI1, the gains are distributed across all four metrics, including a 1.43-point ARI improvement over the strongest displayed comparator. On Mutagenicity, the margins are narrower but remain positive in every cell. On COLLAB and MUTAG, the largest gains again occur in ARI, at 1.91 and 1.78 points, respectively. On PTC_MR, ARET-GCL improves NMI and ARI by 1.63 and 2.33 points while also retaining 0.90-point gains in ACC and macro-F1. The full supplied values in Appendix B show that this conclusion follows directly from the table rather than from an additional project result.

5.2 WHAT THE TABLE VALIDATES

The breadth of the 20 positive margins is consistent with the intended three-stage design. D11-I01 acts before encoding, D2-I09 controls how cross-view structural disagreement enters relation construction, and D10-I10 retains layerwise change in the contrastive coordinates. Their combination produces an endpoint that remains competitive across all five supplied datasets and across predictive and partition-oriented metrics. In this bounded sense, the experiment completes the problem-diagnosis-solution-validation chain: the three diagnosed failure locations motivate three coordinated interventions, and the resulting full system improves every displayed cell.

The same evidence does not identify which intervention produced each gain. The supplied table contains neither isolated modules nor leave-one-module-out variants, so the 20 positive margins cannot be decomposed into three causal effects or used to prove synergy. Moreover, the six comparators were selected under the condition that their means are below ARET-GCL in all 20 cells. The perfect displayed win rate is therefore a property of this comparison panel, not evidence of universal state-of-the-art dominance. The supported claim is narrower and precise: the complete D11-I01 + D2-I09 + D10-I10 system shows consistent cross-dataset and cross-metric improvement over the six selected comparators in the supplied table.

6 DISCUSSION AND LIMITATIONS

The central contribution of ARET-GCL is the organization of three innovations into one reliability chain. D11-I01 addresses the earliest failure point by forcing the two views to experience different structural perturbations. D2-I09 addresses the next failure point by treating cross-view neighborhood-residual disagreement as a reason to discount structural evidence before it shapes routing and hard negatives. D10-I10 addresses the final failure point by replacing an endpoint-only coordinate with a compact contrast of selected encoder states. The design therefore progresses from creating complementary evidence, to judging its structural consistency, to retaining how it evolves.

The supplied table supports the combined endpoint of this chain. Positive margins across all five datasets and all four metrics indicate that the full system’s advantage is not tied to a single benchmark or to accuracy alone. The comparatively large ARI margins on NCI1, COLLAB, MUTAG, and PTC_MR are particularly consistent with the paper’s emphasis on structural partition quality, while the simultaneous ACC and macro-F1 gains show that this emphasis does not require sacrificing the displayed predictive metrics. These interpretations remain descriptive because the table reports aggregate outcomes rather than internal module diagnostics.

Two evidence boundaries are essential. First, the table’s six comparators were selected from a larger evaluated pool under the condition that they remain below our model in all 20 displayed cells. The table is therefore suitable for demonstrating consistency against that selected panel, but not for claiming universal superiority. Second, the table contains no component ablation, alternative ordering, or per-module observable. It validates $D11-I01 + D2-I09 + D10-I10$ as a complete system but cannot establish that each module is independently necessary or that their interaction is synergistic. Such claims require matched module-removal experiments and are outside the evidence provided here.

7 CONCLUSION

We began with a pipeline-level problem: graph contrastive learning can propagate information loss from view construction into relation learning and then hide that loss in endpoint coordinates. We diagnosed three corresponding causes and addressed them with three innovations: asymmetric view construction in $D11-I01$, residual-disagreement trust gating in $D2-I09$, and layer-trajectory coordinate compression in $D10-I10$. The supplied five-dataset table provides the final link in this story. The complete system achieves the largest reported mean in every one of the 20 ACC, NMI, ARI, and macro-F1 cells relative to the six selected comparators, with margins from 0.26 to 2.33 percentage points. This evidence supports consistent end-to-end performance within the supplied comparison and motivates future module-wise validation without overstating what the current table can prove.

AI USE STATEMENT

Generative AI tools assisted manuscript organization, prose editing, LaTeX preparation, and consistency checks. They did not run experiments or create missing measurements. Every empirical value and comparison reported in the paper is restricted to the supplied `table.md`; the authors remain responsible for verifying the final manuscript.

REPRODUCIBILITY STATEMENT

The paper states the three-module dataflow and equations needed to identify the proposed design. Appendix B reproduces the complete experimental evidence used by the manuscript. No result outside that supplied table is used to support an empirical claim.

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A THE THREE INNOVATION POINTS

Table 2 makes the one-to-one correspondence between the diagnosed failure chain and the three proposed modules explicit. These are the only modules claimed as innovations in this paper.

B COMPLETE SUPPLIED EXPERIMENTAL TABLE

The following values reproduce the complete experimental content of the supplied `table.md`. Bold marks the largest mean within each dataset–metric column. The six comparators are the strongest methods, by the mean over all 20 dataset–metric cells, among the 35 evaluated adaptations that remain strictly below ARET-GCL in every displayed cell.

Table 2: Problem–diagnosis–solution mapping for the three innovations.

Module	Diagnosed failure	Implemented innovation	Role in the chain
D11-I01	Paired views may repeat the same destructive perturbation pattern	Asymmetric node-drop and connected-subgraph view construction	Create complementary structural challenges before encoding
D2-I09	Cross-view structural disagreement may enter routing and negative construction as if it were reliable	Batch-relative neighborhood-residual disagreement with sigmoid gating	Discount inconsistent structural evidence before relation construction
D10-I10	A final-state projection may erase how evidence changes across encoder depth	Three-state algebraic second contrast with a learned bottleneck	Retain a compact layer-trajectory signature in the contrastive coordinates

Table 3: Supplied comparison (mean $\pm$ standard deviation, %).

Method	ACC	NMI	ARI	Macro-F1
NCII				
TOPER	74.09 $\pm$ 2.74	18.37 $\pm$ 4.34	23.36 $\pm$ 5.31	73.94 $\pm$ 2.77
Khan-GCL	73.65 $\pm$ 2.58	17.31 $\pm$ 3.99	22.49 $\pm$ 4.78	73.59 $\pm$ 2.55
HSPGKN	73.16 $\pm$ 1.72	16.64 $\pm$ 2.84	21.46 $\pm$ 3.31	73.05 $\pm$ 1.70
DualPrism	70.56 $\pm$ 2.54	12.90 $\pm$ 3.16	17.01 $\pm$ 4.18	70.50 $\pm$ 2.56
KAGNN	73.72 $\pm$ 0.94	17.07 $\pm$ 1.37	22.44 $\pm$ 1.81	73.69 $\pm$ 0.96
DGDA	72.44 $\pm$ 3.53	15.87 $\pm$ 4.67	20.44 $\pm$ 6.15	72.29 $\pm$ 3.62
D11-I01 + D2-I09 + D10-I10	74.87 $\pm$ 2.17	19.01 $\pm$ 3.53	24.79 $\pm$ 4.33	74.83 $\pm$ 2.17
Mutagenicity				
TOPER	78.81 $\pm$ 2.39	25.35 $\pm$ 4.31	33.29 $\pm$ 5.35	78.40 $\pm$ 2.46
Khan-GCL	77.56 $\pm$ 1.18	22.85 $\pm$ 2.16	30.34 $\pm$ 2.67	77.24 $\pm$ 1.22
HSPGKN	77.61 $\pm$ 2.49	23.33 $\pm$ 4.13	30.57 $\pm$ 5.52	76.96 $\pm$ 2.94
DualPrism	78.69 $\pm$ 1.85	25.17 $\pm$ 3.57	32.94 $\pm$ 4.25	78.22 $\pm$ 2.05
KAGNN	78.76 $\pm$ 3.51	25.54 $\pm$ 6.64	33.39 $\pm$ 8.07	78.34 $\pm$ 3.79
DGDA	77.15 $\pm$ 2.74	22.38 $\pm$ 4.84	29.62 $\pm$ 6.07	76.65 $\pm$ 2.87
D11-I01 + D2-I09 + D10-I10	79.13 $\pm$ 1.83	25.80 $\pm$ 3.65	33.96 $\pm$ 4.36	78.81 $\pm$ 1.91
COLLAB				
TOPER	72.08 $\pm$ 1.65	28.70 $\pm$ 2.24	31.39 $\pm$ 3.03	69.26 $\pm$ 2.72
Khan-GCL	72.72 $\pm$ 1.98	27.92 $\pm$ 3.15	32.59 $\pm$ 3.88	69.68 $\pm$ 2.03
HSPGKN	69.88 $\pm$ 1.84	26.96 $\pm$ 3.11	27.33 $\pm$ 3.22	65.96 $\pm$ 0.79
DualPrism	70.26 $\pm$ 1.85	26.09 $\pm$ 2.79	27.75 $\pm$ 3.39	67.48 $\pm$ 3.04
KAGNN	72.50 $\pm$ 1.56	28.60 $\pm$ 2.17	31.53 $\pm$ 2.86	69.67 $\pm$ 1.68
DGDA	70.06 $\pm$ 1.34	27.89 $\pm$ 1.17	27.77 $\pm$ 1.85	65.78 $\pm$ 1.01
D11-I01 + D2-I09 + D10-I10	73.62 $\pm$ 1.76	29.14 $\pm$ 2.97	34.50 $\pm$ 3.76	70.69 $\pm$ 1.61
MUTAG				
TOPER	68.15 $\pm$ 7.69	12.70 $\pm$ 14.09	9.06 $\pm$ 15.94	57.28 $\pm$ 17.34
Khan-GCL	85.62 $\pm$ 4.10	42.61 $\pm$ 12.94	49.64 $\pm$ 12.25	84.11 $\pm$ 4.02
HSPGKN	77.18 $\pm$ 6.94	25.91 $\pm$ 17.78	26.78 $\pm$ 17.06	68.67 $\pm$ 17.49
DualPrism	72.32 $\pm$ 6.96	12.89 $\pm$ 13.65	15.87 $\pm$ 15.83	61.52 $\pm$ 16.05
KAGNN	68.63 $\pm$ 6.18	5.19 $\pm$ 7.37	7.25 $\pm$ 11.72	51.37 $\pm$ 14.40
DGDA	72.82 $\pm$ 8.85	15.99 $\pm$ 24.14	15.06 $\pm$ 23.35	54.52 $\pm$ 21.02
D11-I01 + D2-I09 + D10-I10	86.13 $\pm$ 5.23	43.38 $\pm$ 18.28	51.42 $\pm$ 16.79	84.48 $\pm$ 6.28
PTC_MR				
TOPER	58.13 $\pm$ 2.58	1.46 $\pm$ 1.76	1.48 $\pm$ 1.79	47.12 $\pm$ 9.10
Khan-GCL	60.76 $\pm$ 3.79	3.44 $\pm$ 2.39	3.70 $\pm$ 3.25	60.07 $\pm$ 3.86
HSPGKN	56.95 $\pm$ 3.67	1.87 $\pm$ 2.43	0.98 $\pm$ 2.52	47.15 $\pm$ 8.99
DualPrism	54.96 $\pm$ 1.97	0.09 $\pm$ 0.18	-0.27 $\pm$ 0.60	39.39 $\pm$ 6.49
KAGNN	56.97 $\pm$ 1.58	1.73 $\pm$ 2.64	0.67 $\pm$ 0.93	39.58 $\pm$ 4.07
DGDA	52.04 $\pm$ 6.59	0.44 $\pm$ 0.72	0.18 $\pm$ 0.85	39.27 $\pm$ 7.42
D11-I01 + D2-I09 + D10-I10	61.66 $\pm$ 7.80	5.07 $\pm$ 4.88	6.03 $\pm$ 6.97	60.97 $\pm$ 7.52